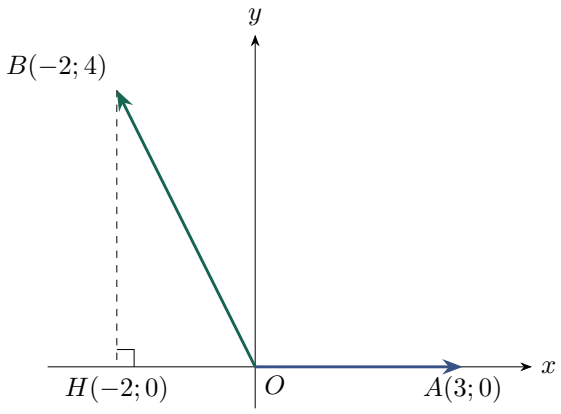
Projection and angle

Definition and results

For two non-zero vectors $\vec{u}, \vec{v}$, their scalar product is real $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$where: $\theta \in [0; \pi]$ is their angle not oriented. With a zero vector, the product is 0. If H is the projected of B on (OA), $\overrightarrow{OA} \cdot \overrightarrow{OB}$ value $OA \times OH$ when H is on the half-right [OA) and its opposite If not.

Status and justification. Definition; equivalence with projection results from cosine in the Rectangular triangles, with negative algebraic projection for an obtuse angle.

Example solved. For two vectors of norms 3 and 4 forming 60 °the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120 °, it's worth -6. If $A(3; 0)$ and $B(-2; 4)$, the projected is $H(-2; 0)$ and the product is -6.



Projected H is on the negative side of the horizontal axis. $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

Point of vigilance

The result is a number, not a vector. Right angle is 0, not 1.

Immediate application. Two vectors have norms 2 and 5 and form 90 °, then 180 °. Calculate products.

3.2 Coordinates, Norm and Orthogonality

Definition and results

In an orthonormous base, $\vec{u} = (x; y)$ and $\vec{v} = (x\prime; y\prime)$, $\vec{u} \cdot \vec{v} = xx\prime + yy\prime$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their product is zero. For a base ($i, \vec{j}$) orthonormous, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

Status and justification. Demonstrated properties using bilinearity: develop ($x\vec{i} + y\vec{j}$) · ($x\prime\vec{i} + y\prime\vec{j}$) and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

Example solved. $\vec{u} = (2; 3)$, $\vec{v} = (3; -2)$: output 6 − 6 = 0. Both vectors are orthogonal. Their norm is $\sqrt{13}$. The product of (2; 3) with (1; 0) is worth 2: we find the abscisse.

Point of vigilance

The usual formula assumes an orthonormous basis. different properties.

Immediate application. For what real t the vectors (t; (2) and (3); -6) Are they orthogonal?

3.3 Vector calculations and identities

Definition and results

The scalar product is symmetrical and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. So $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. We're getting it. $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

Status and justification. Bilinearity accepted from the geometric definition; symmetry resulting Cosine. Identities are demonstrated by developing ($\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

Example solved. If $\|\vec{u}\| = 3$, $\|\vec{v}\|$ = 4 and $\|\vec{u} + \vec{v}\|$ = 5, the product is $\dfrac{25 - 9 - 16}{2}$ = 0. Deduct orthogonality without coordinates.

Point of vigilance

In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. Identities concern the squares of norms.

Immediate application. We give $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v}$ = 1. Calculate $\|\vec{u} - \vec{v}\|$.

3.4 Al-Kashi and choice of method

Definition and results

In an ABC triangle, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows calculate the third side from two sides and including the angle, or determine the cosine d-an angle from the three sides.

Status and justification. By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Developing its scalar square gives $BC^2 = AC^2 + AB^2 - 2AB \cdot AC$; introduce the expression with the cosine.

Example solved. $AB = 5$, AC = 3 and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15$ = 19, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approaching the angle to the calculator if allowed.

Point of vigilance

The angle is between the two sides of the product term. The minus sign is essential.

Immediate application. A triangle a $AB = AC$ = 4 and $BC = 4\sqrt{2}$. Determine angle A.

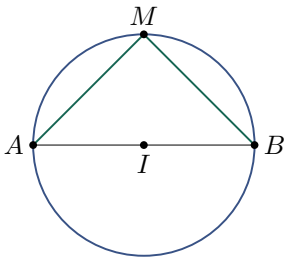
3.5. The AB diameter circle

Definition and results

For two separate points A and B of midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. All points M such as $\overrightarrow{MA} \cdot \overrightarrow{MB}$ = 0 is the centre circle I and radius circle $\dfrac{AB}{2}$, points A and B included.

Status and justification. Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. Cross-words cancel ; it remains $MI^2 - IA^2$. Like $IA = \dfrac{AB}{2}$, equal to zero is equivalent to the metric equation of the circle.

Example solved. $A(-2; 0)$ and $B(2; 0)$: I is the origin. The locus is $x^2 + y^2$ = 4. For $M(0; 1)$, the product is 1 $- 4 = -3$; for $M(0; 2)$, it's 0.



AB diameter circle: vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

Point of vigilance

The ends belong to the place even if "right triangle" is not suitable for a Degenerate triangle.

Immediate application. For $A(0;0)$, $B(6;0)$, determine M such as $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.